4-Day Machine Learning Crash Plan (with Preprocessing)

Day 1: ML Basics & Regression

- ML Introduction: Supervised vs Unsupervised Learning
- Setup: scikit-learn, Jupyter Notebook or Google Colab
- Load dataset (e.g., Iris or Boston Housing)
- Preprocessing:
  - Handle missing values (if any)
  - Standardization: Use StandardScaler from sklearn
- Build and train a Linear Regression model
- Evaluate using Mean Squared Error (MSE)

Keywords: train_test_split, LinearRegression, StandardScaler

Day 2: Classification + Label Encoding

- Use Iris dataset for classification
- Preprocessing:
  - Label Encoding: Encode categorical labels using LabelEncoder
  - Normalization: Use MinMaxScaler to scale features
- Apply Logistic Regression model
- Evaluate with Accuracy Score and Confusion Matrix

Keywords: LabelEncoder, MinMaxScaler, LogisticRegression, confusion_matrix

Day 3: Clustering & Visualization

- Apply K-Means clustering on Iris (without labels)
- Preprocessing: Standardize features using StandardScaler

- Visualize clusters using matplotlib or seaborn

Keywords: KMeans, fit_predict, scatter, StandardScaler

Day 4: Mini Project + Revision

- Choose a small dataset from sklearn or Kaggle
- Preprocessing workflow:
  - Label Encode if needed
  - Apply StandardScaler or MinMaxScaler
- Train a Regression or Classification model
- Evaluate results and visualize insights

Final Checklist:

- [ ] Label Encoding
- [ ] StandardScaler
- [ ] MinMaxScaler
- [ ] Linear Regression
- [ ] Logistic Regression
- [ ] KMeans Clustering
- [ ] Confusion Matrix & Accuracy